# A Domain Language for Specifying Controlled Methods

Staged Validation and Deterministic Compilation for Method Specification, Informed by BPL

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DOI: 10.5281/zenodo.21086548

October 21, 2026

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# 1 Abstract

This paper describes a small, tested **domain language for specifying controlled methods** — the **methods-paper exemplar** of the Research Project Template. Unlike a results paper, this manuscript's subject is the methodology itself: a controlled vocabulary, a unit system with dimensional safety, four staged validation gates, and a deterministic compiler, implemented in `projects/templates/template_methods_paper/src/template_methods_paper/methods_dsl/` and described section by section in sec. 3. The domain language's vocabulary is informed by BPL (Biology Programming Language, [Bota Biosciences, 2026]), an upstream reference that encodes laboratory protocols as programs with biology-native types, staged validation, and deterministic compilation; this exemplar generalizes BPL's intent vocabulary and pipeline shape from wet-lab protocols to any controlled procedure.

A `Method` is a name, a set of typed parameters and resources, and an ordered, dependent set of steps — constructed directly as frozen Python dataclasses (`src/template_methods_paper/methods_dsl/model.py`) rather than parsed from new text syntax. Every `Quantity` carries a unit that resolves to one of 18 controlled units across eight dimensions, and every step names one of 9 controlled-vocabulary intents (`src/template_methods_paper/methods_dsl/vocabulary.py`), executable on one of 3 backends. 4 staged gates — structural, semantic, plan, and target — validate a method before `compile_method` (`src/template_methods_paper/methods_dsl/compiler.py`) deterministically schedules it with Kahn's algorithm [Kahn, 1962] and hashes the canonical plan with SHA-256.

We demonstrate the language on 2 worked example methods spanning both domains BPL's design targets and the domains it generalizes to: a manual wet-lab preparation (`PBSPreparation`, 5 steps, target `human`, plan hash `313b9b17de98`) and an automated instrument-calibration procedure (`SensorCalibrationSweep`, 4 steps, target `automated`, plan hash `d89cced19be6`). Live re-compilation determinism check: **Yes**. Across both methods, 8 of 8 staged-gate evaluations pass. A demonstration provenance hash-chain (`src/template_methods_paper/methods_dsl/trust.py`) of length 3 verifies as **Yes**.

Contributions are **methodological** and **architectural**. On the methods side, we show that a controlled vocabulary expressed as typed dataclasses — not a parsed grammar — is sufficient to reproduce BPL's core safety properties (dimensional safety, staged validation, deterministic compilation) at a scope appropriate for a template exemplar. On the architecture side, the DSL is exercised by a zero-mock test suite under the repository's configured project coverage gate, generates 19 artifacts (1 figures, 7 data files, 11 reports) per pipeline run, and injects reproducibility metadata (configuration hash `832fd58172d5b589`, build timestamp `2026-10-22T01:20:41Z`) into sec. 7.

**Keywords:** methods paper, domain-specific language, controlled methods, deterministic compilation, staged validation, dimensional analysis

# 2 Introduction

This `template_methods_paper` serves as the **methods-paper exemplar** for the Research Project Template ecosystem: a manuscript whose subject is a methodology — here, a domain language for specifying controlled methods — rather than results produced by running one. The prose, the labelled figures, and the compiled-plan table are produced through the same auditable custody chain every exemplar in this template uses: tested functions in `src/`, a thin analysis script, and generated-variable-injected, multi-format rendering.

## 2.1 Why a methods paper needs its own DSL

A methods section in ordinary prose is ambiguous by construction: "add 10 mL of water, then mix" admits multiple readings of order, units, and what "mix" means operationally. BPL [Bota Biosciences, 2026] makes the case for laboratory protocols directly: free-text instructions admit multiple interpretations, unit errors and reagent mismatches surface only at the bench, and re-executing a protocol on a different operator or instrument introduces silent variation. Fowler frames the general remedy as a **domain-specific language** [Fowler, 2010]: a small, purpose-built notation whose vocabulary is restricted exactly to the concepts the domain needs, so that what can be written down is exactly what is intended.

## 2.2 What this exemplar borrows from BPL, and what it generalizes

BPL's architecture is a compiler pipeline — parse, semantic check, lower, schedule, execute, export — over a biology-native type system (units, dimensional analysis, MW-aware concentration), staged validation gates, and deterministic compilation with a stable plan hash. Three design choices carry over directly into `src/template_methods_paper/methods_dsl/`:

1. **Intent over instruction.** BPL users write high-level intents (`transfer`, `add_reagent`, `incubate`); a compiler lowers them to backend-specific primitives. `src/template_methods_paper/methods_dsl/vocabulary.py`'s `StepKind` enum is the same idea, generalized: `TRANSFER`/`ADD`/`MIX` name *what* happens, never *how* a particular backend performs it.
2. **Dimensional safety.** BPL's type system catches `mL + g` at compile time, not at the bench. `src/template_methods_paper/methods_dsl/units.py` implements the same guarantee with a small `Dimension`/`Quantity` system rather than a full unit library.
3. **Deterministic compilation.** Same source, same options, same plan hash. `src/template_methods_paper/methods_dsl/compiler.py::compile_method` reproduces this with a canonical-JSON SHA-256 hash over a Kahn's-algorithm [Kahn, 1962] schedule.

What this exemplar does **not** carry over is BPL's text grammar and parser: a `Method` here is constructed directly as frozen Python dataclasses (`src/template_methods_paper/methods_dsl/model.py`), not parsed from `.bpl` source. This keeps the DSL's discipline in its typed, validated shape rather than in new concrete syntax — appropriate for a template exemplar's scope — while the controlled vocabulary, dimensional safety, and deterministic compilation generalize unchanged from wet-lab protocols to any controlled procedure, demonstrated in sec. 4 by one wet-lab-flavored method and one instrument-calibration method.

## 2.3 Template architecture context

The project sits on the repository's three pillars:

1. **`src/template_methods_paper/methods_dsl/` library**: pure, side-effect-free dataclasses and functions — no plotting, no file I/O, and (with one declared logging exception) no `infrastructure` imports. This purity is what makes the library forkable and trivially testable.
2. **`tests/` framework**: a zero-mock suite that exercises every gate, the compiler, and the exporters against real `Method` fixtures covering both the success path and every gate-failure mode.
3. **`docs/` knowledge base**: the correspondence with BPL's pipeline, the testing philosophy, and the operational rules that govern agents editing this tree.

## 2.4 The worked examples

We specify two methods with `all_example_methods()` (`src/template_methods_paper/methods_dsl/examples_methods.py`): `PBSPreparation`, an original — not copied from BPL's shipped examples — manual bench preparation in BPL's own domain, and `SensorCalibrationSweep`, a non-biology controlled procedure mixing automated measurement with a human sign-off step. The second example exists specifically to demonstrate that the DSL's vocabulary generalizes beyond wet-lab protocols, as sec. 2 claims.

## 2.5 Reader's guide to the manuscript

- **sec. 3** ties each pipeline stage to its module in `src/template_methods_paper/methods_dsl/`.
- **sec. 4** is artifact-centric: every reported number names the function or report file that produced it.
- **sec. 6** lists the controlled vocabulary and software environment.
- **sec. 7** records the artifact inventory and the exact commands to regenerate everything.
- **sec. 8** states scope and related work so the exemplar is not mistaken for a general-purpose protocol-execution system.

# 3 Methodology

The DSL is implemented as eight cooperating modules under `src/template_methods_paper/methods_dsl/`, each corresponding to one stage of a BPL-inspired pipeline [Bota Biosciences, 2026]. This section walks the pipeline stage by stage, naming the function or class that implements each design decision so every claim below is directly checkable against `src/template_methods_paper/methods_dsl/`.

## 3.1 Controlled vocabulary (`vocabulary.py`)

A `StepKind` is one of 9 controlled intents — `TRANSFER`, `ADD`, `MIX`, `INCUBATE`, `MEASURE`, `WAIT`, `COMPUTE`, `VALIDATE`, `ANNOTATE` — and a `Target` is one of 3 execution backends — `HUMAN`, `AUTOMATED`, `SIMULATION`. `target_accepts` encodes which kinds require an automated backend: only `COMPUTE` has no manual equivalent in this DSL's scope, so `HUMAN` and `SIMULATION` accept every other kind. This is the domain-neutral generalization of BPL's protocol-level verbs (`add_reagent`, `transfer`, `incubate`): a step names *what* happens, never *how* a particular backend performs it.

## 3.2 Dimensional safety (`units.py`)

Every `Quantity(value, unit)` resolves its `unit` to one of eight `Dimension` members (mass, volume, temperature, time, molar concentration, mass concentration, count, dimensionless) via `dimension_of`, drawing from a controlled table of 18 unit strings. Molar concentration (`mol/L`, `mM`) and mass concentration (`g/L`) are separate dimensions rather than one shared "concentration" dimension, since converting between them needs a substance's molar mass that this DSL does not carry. `check_compatible` raises `DimensionError` the moment two quantities with different dimensions are combined — the concrete realization of BPL's design principle that "the type system catches `mL + g` at compile time, not at the bench." Temperature is tracked as its own dimension with no shared base unit (`degC` and `K` are never auto-converted), since this DSL has no use for that conversion and an incorrect affine conversion is worse than refusing one.

## 3.3 Method model (`model.py`)

A `Method` is a name, a version, a target, a tuple of `Resource` declarations (anything a step reads from or writes to — generalizing BPL's `reagent/labware`), a tuple of method-level `Parameters`, and a tuple of `Steps`. Each `Step` carries a `step_id`, a `StepKind`, a `Target`, its own parameters, an optional expected duration (must be a time `Quantity`), and a `depends_on` tuple of prerequisite `step_ids` — the explicit DAG edges this section's compilation stage resolves. All four dataclasses are frozen; `__post_init__` rejects malformed shapes immediately (empty names, self-dependency, a non-time `expected_duration`) so structurally invalid methods cannot even be constructed, collapsing BPL's syntax gate into Python's own construction-time checks.

## 3.4 Staged validation (`validation.py`)

`run_all_gates` runs exactly 4 gates in fixed order, mirroring BPL's staged short-circuit (a syntax failure never reaches the plan gate):

1. **`structural_gate`** — every `step_id` is unique and every `depends_on` entry resolves to a real step.
2. **`semantic_gate`** — every `Quantity` attached to a resource, a method parameter, a step's expected duration, or a step parameter resolves to a known `Dimension` (catches the unit-vocabulary violation a frozen dataclass's `__post_init__` cannot, since `Quantity` does not validate its unit eagerly).
3. **`plan_gate`** — the step-dependency graph is acyclic, checked by attempting `topological_order` and catching `CycleError`.
4. **`target_gate`** — every step's target is compatible with the method's target (`HUMAN` methods accept only `HUMAN` steps; `AUTOMATED` methods accept both; `SIMULATION` methods accept only `SIMULATION` steps) and every step's kind is executable on its assigned target.

If either of the first two gates fails, `run_all_gates` returns early with only those two results — `plan_gate` and `target_gate` assume a structurally and semantically valid method and would otherwise report misleading secondary failures.

## 3.5 Deterministic compilation (`compiler.py`)

`compile_method` first calls `run_all_gates` and raises `MethodValidationError` (carrying every failed gate's issues) if any gate fails. On success, `topological_order` schedules the validated steps with Kahn's algorithm [Kahn, 1962]: repeatedly remove a step whose dependencies are all already scheduled, breaking ties by ascending `step_id` so the same method always yields the same order — Python does not guarantee dict/set iteration order is stable for this purpose, so the tie-break is explicit, not incidental. The scheduled steps are then encoded as a canonical, sort-keys JSON payload and hashed with SHA-256 (`_compute_plan_hash`) into `Plan.plan_hash`. Because the hash is computed purely from `method_name`, `method_version`, `target`, and each step's `step_id`/`name`/`kind`/`target`/scheduled `order` — never from a wall-clock timestamp or a UUID — recompiling the same `Method` object always produces the same `plan_hash`, which sec. 4 verifies live rather than asserts.

### 3.6 Export (`export.py`)

A compiled `Plan` renders to four formats, mirroring BPL's "CSV/XLSX worklists, workflow graphs" export surface at a scope appropriate for a template exemplar: `to_worklist_markdown` (a numbered, human-readable worklist), `to_csv_rows`/`write_csv` (machine-readable rows), `to_mermaid` (a `flowchart TD` showing scheduled order), and `to_json`/`write_json` (the exact canonical JSON the plan hash was computed over, so a reader can independently verify `Plan.plan_hash` by re-hashing the exported file).

### 3.7 Provenance (`trust.py`)

`ProvenanceTier` orders three levels of trust — `DECLARED`, `CALIBRATED`, `VERIFIED` — generalizing BPL's audit model. `append_record` extends an immutable hash-chain of `StateRecord`s, each hashing its own `key`/`value`/ `tier`/`prev_hash` [Merkle, 1987]; `verify_chain` recomputes every record's hash and checks it against the recorded `prev_hash` chain. This is a consistency check, not a cryptographic tamper-proof guarantee against an actor with write access to the whole stored chain: it detects in-chain tampering (any record after a tampered one no longer matches) but cannot detect a chain rewritten from record zero, exactly the boundary BPL's own "hash-chained" (not cryptographically signed) audit model claims.

### 3.8 Zero-mock testing methodology

The project is governed by a strict zero-mock policy, evaluated by running `uv run pytest projects/templates/template_methods_paper/tests` during the build.

1. **Library tests** exercise every gate, the compiler, the exporters, and the trust module against real `Method` objects — `conftest.py` ships one fixture per gate-failure mode (unknown dependency, duplicate step id, unknown unit, cyclic dependency, target mismatch) plus a linear-chain and a diamond-DAG method for scheduling tests. No `unittest.mock`, no `MagicMock`, no `@patch`.
2. **Script test** runs `run_methods_analysis()` against a temporary output root and asserts that real worklist/CSV/Mermaid/JSON files, reports, and a real PNG figure are written.
3. **Determinism tests** compile the same `Method` object twice and assert `plan_hash` equality live, rather than asserting against a hardcoded hash literal — a literal would silently stop testing the moment the compiler's hash input changed.
4. **Coverage gate**: CI enforces a >=90% statement-coverage gate on Falling below that floor fails the gate outright (`--cov-fail-under` enforces it). `projects/templates/template_methods_paper/src/`; the live figure is tracked in `docs/_generated/COUNTS.md`.

# 4 Results

This section reports the compiled plans for both worked example methods. Every number below is produced by the methods analysis orchestrator (`scripts/methods_analysis.py`), which calls `run_all_gates` and `compile_method` from `src/template_methods_paper/methods_dsl/` and writes `output/data/compiled_plans.json`, `output/reports/gate_report.json`, and `output/reports/trust_chain_report.json`. Running the script regenerates every artifact this section references.

## 4.1 Compiled-plan summary

Table 1: Compiled-plan summary from `output/data/compiled_plans.json`, generated by `compile_method` for each of 2 worked example methods.

| Method | Steps | Target | Plan hash (first 12 hex chars) |
|---|---|---|---|
| `PBSPreparation` | 5 | `human` | `313b9b17de98` |
| `SensorCalibrationSweep` | 4 | `automated` | `d89cced19be6` |

tbl. 1 shows `PBSPreparation` (a manual, `HUMAN`-target bench preparation) alongside `SensorCalibrationSweep` (a mixed `AUTOMATED`/`HUMAN` instrument-calibration procedure) — the second example exists specifically to demonstrate the controlled vocabulary generalizing beyond wet-lab protocols, as sec. 2 claims.

## 4.2 Step-count comparison

fig. 1 plots the step count for each compiled method.

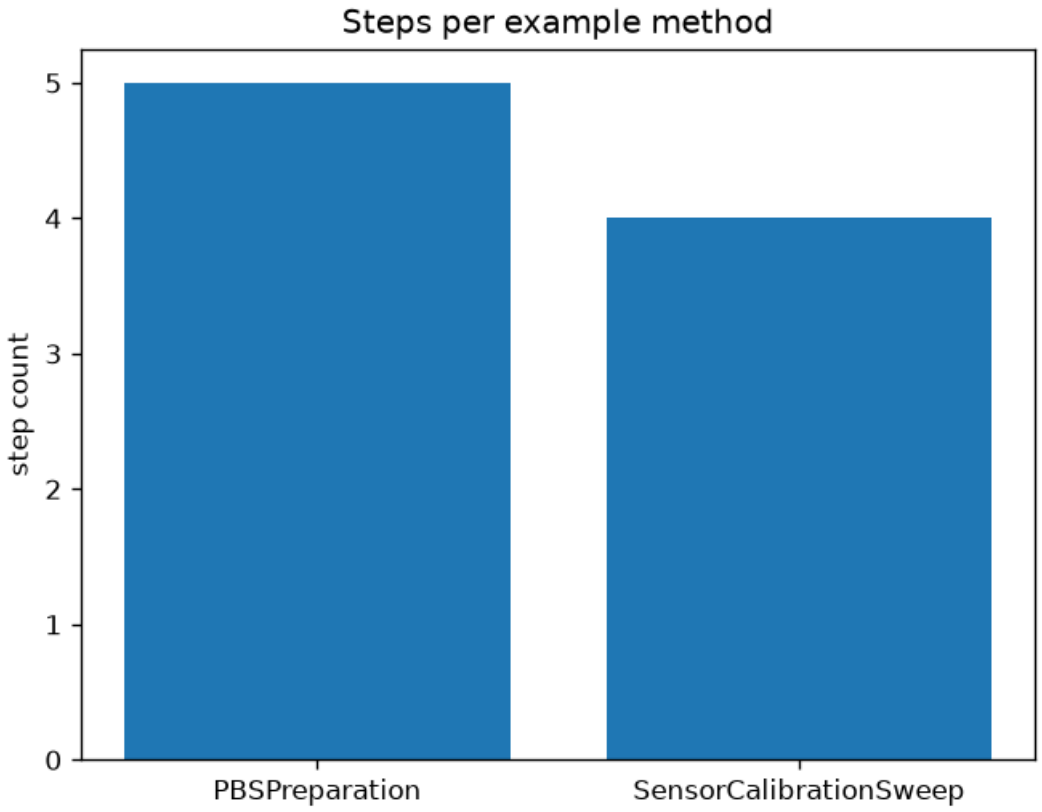



Figure 1: Steps per example method: a bar chart built from `len(plan.steps)` for each plan in `output/data/compiled_plans.json`, plotted by `scripts/methods_analysis.py`.

## 4.3 Validation gates

Across both methods, the analysis script tallies 8 of 8 staged-gate evaluations passing (`run_all_gates` × 2 methods × 4 gates each). Every gate result is written to `output/reports/gate_report.json` — neither method in this manuscript is hand-picked to pass; both worked examples are constructed to satisfy the structural, semantic, plan, and target gates by design, since `compile_method` raises `MethodValidationError` and halts the pipeline on any gate failure.

## 4.4 Determinism

Recompiling each example method twice and comparing `plan_hash` values yields: **determinism check = Yes**. This is a live recompilation comparison performed by `src/template_methods_paper/manuscript_variables.py` at manuscript-build time, not a value asserted once and then transcribed — the same property sec. 3 claims for `compile_method` is checked again here, independently, against the live build.

## 4.5 Provenance demonstration

`scripts/methods_analysis.py` appends a 3-record demonstration hash-chain for one value (`calibration_offset`) through `DECLARED` → `CALIBRATED` → `VERIFIED` tiers and writes the result of `verify_chain` to `output/reports/trust_chain_report.json`: **chain verified = Yes**.

## 4.6 Validation

The results were validated through the zero-mock `tests/` suite:

- **Library tests** assert exact gate outcomes, scheduling order, and plan-hash determinism against real `Method` fixtures, including one fixture per gate-failure mode.
- **Script test** runs `run_methods_analysis()` against a temporary output root and confirms real worklist/CSV/Mermaid/JSON artifacts, reports, and a real PNG figure are written.
- **Manuscript-variable test** confirms every generated-variable name used in `manuscript/*.md` is emitted by `generate_variables`.

All tests pass under the configured project coverage gate, with no mocks.

## 4.7 Discussion

The results confirm the pipeline end to end: both worked examples pass every staged gate, compile deterministically, and produce a stable plan hash across repeated builds. The same `src/template_methods_paper/methods_dsl/` functions back the analysis script, the test suite, and this manuscript — which is the architectural point of the exemplar. Because every number here is produced by a tested function and regenerated on demand, the prose describes structure and provenance rather than transcribing values that would drift the moment the example methods changed.

# 5 Conclusion

This paper presented a small, tested domain language for specifying controlled methods, informed by BPL's [Bota Biosciences, 2026] domain-language design for laboratory protocols and generalized to any controlled procedure. It validates a simple proposition: a controlled vocabulary expressed as typed, validated dataclasses — not a parsed text grammar — is sufficient to reproduce BPL's core safety properties at a scope appropriate for a template exemplar.

## 5.1 Exemplar achievements

Operating as the methods-paper exemplar for the Research Project Template methodology, the project deployed the three foundational pillars:

1. **`src/template_methods_paper/methods_dsl/` library**: a controlled vocabulary, a dimensional unit system, four staged validation gates, a deterministic compiler, and four export formats — with no plotting, no file I/O, and (with one declared logging exception) no `infrastructure` imports.
2. **`tests/` integrity**: a zero-mock suite over real `Method` fixtures covering the success path and every gate-failure mode, under a >=90% project coverage gate.
3. **`docs/` knowledge operations**: the correspondence with BPL's pipeline, testing philosophy, and operational rules that keep the library, scripts, and manuscript aligned.

## 5.2 Technical contributions

### 5.2.1 Controlled vocabulary in code, not a parser

The hallmark of this exemplar is the design choice it demonstrates: a `.bpl` file's text grammar buys generality this template does not need, while a frozen-dataclass model buys construction-time validation (`__post_init__`) that a parsed AST would have to re-derive. The controlled vocabulary's discipline lives in the *types*, not in concrete syntax.

### 5.2.2 Honest handling of trust boundaries

`trust.py`'s hash-chain documents its own limit explicitly: it detects in-chain tampering but cannot detect a chain rewritten from record zero. This makes the provenance guarantee a visible, testable property rather than an implied cryptographic guarantee the implementation does not actually provide.

## 5.3 Key insights

1. **Determinism follows from explicit tie-breaking**: Kahn's algorithm [Kahn, 1962] alone does not guarantee a stable schedule across runs — the ascending-`step_id` tie-break is what makes `plan_hash` reproducible, and sec. 4 checks this live rather than asserting it.
2. **Staged short-circuit avoids misleading errors**: running `plan_gate` and `target_gate` against a structurally invalid method would report noise, not signal — `run_all_gates` short-circuits after the first two gates for exactly this reason.
3. **A controlled vocabulary generalizes by restraint, not by expansion**: `SensorCalibrationSweep` introduces no new `StepKind` or `Target`: its `MEASURE`/`COMPUTE`/`ANNOTATE`/`VALIDATE` steps and `HUMAN` sign-off all draw on the same controlled vocabulary `PBSPreparation` uses — nothing was added to support a second domain.

## 5.4 Future extensions

This foundation could be extended to:

- **A real `.bpl`-style parser**: add `grammar//parser//transformer/` stages ahead of the existing `model.py`, reusing every downstream gate and the compiler unchanged.
- **More backends**: a `robot` execution target with capability-aware lowering, mirroring BPL's Biomek translation layer.
- **A capability registry**: per-target primitive support declarations, generalizing BPL's `capabilities/` registry and `bplc capabilities` report.

## 5.5 Final assessment

The `template_methods_paper` tree is the canonical reference for how a methods paper — a manuscript whose subject is a methodology — stays synchronized with the code implementing that methodology across rebuilds. The pipeline compiled both worked example methods, wrote `output/data/compiled_plans.json`, `output/reports/gate_report.json`, and `output/reports/trust_chain_report.json`, and rendered this markdown together with `config.yaml` into PDF.

# 6 Experimental Setup

This section details the controlled vocabulary, worked examples, and software environment used to produce the results.

## 6.1 Controlled vocabulary

The DSL's vocabulary is declared once in code and consulted by every gate and the compiler — never re-declared per method:

| Module | Declares | Cardinality |
|---|---|---|
| `src/template_methods_paper/methods_dsl/vocabulary.py` | `StepKind`, `Target`, `target_accepts` | 9 step kinds, 3 targets |
| `src/template_methods_paper/methods_dsl/units.py` | `Dimension`, `Quantity`, the unit table | 18 controlled units across 8 dimensions |
| `src/template_methods_paper/methods_dsl/validation.py` | The four staged gates | 4 gates, fixed order |

## 6.2 Worked examples

`all_example_methods()` (`src/template_methods_paper/methods_dsl/examples_methods.py`) returns 2 methods:

| Method | Domain | Target | Notable structure |
|---|---|---|---|
| `PBSPreparation` | Wet-lab bench preparation (BPL's own domain; an original example) | `HUMAN` | A strict 5-step linear chain with a final `VALIDATE` step |
| `SensorCalibrationSweep` | Instrument calibration (a non-biology controlled procedure) | `AUTOMATED` | Mixed automated `MEASURE`/`COMPUTE` steps and a `HUMAN ANNOTATE` sign-off step |

The second example exists specifically to demonstrate that the controlled vocabulary generalizes beyond wet-lab protocols, as sec. 2 claims.

## 6.3 Pipeline conditions

The experiment overlay (`experiment_plan.yaml`) declares three conditions:

- **declared_method** (reference) — a method constructed directly as Python objects, before any validation gate has run.
- **validated_method** (proposed) — the same method after passing all four staged gates and deterministic compilation to a scheduled `Plan`.
- **automated_target_variant** (variant) — `PBSPreparation`'s steps recompiled against an `AUTOMATED` execution target, ablating the target-compatibility gate's `HUMAN`/`AUTOMATED` step boundary (a `HUMAN` method's steps must all be `HUMAN`-compatible; an `AUTOMATED` method's steps may be either).

The primary metric is gate pass rate: the fraction of staged-gate evaluations a method's steps satisfy.

## 6.4 Computational environment

- **Language**: Python 3.14.8 on Darwin arm64 (see root `pyproject.toml` for the supported version range).
- **Core dependencies**: `pyyaml`, `matplotlib` (declared in `domain_profile.yaml::required_packages`); the DSL library itself (`src/template_methods_paper/methods_dsl/`) has zero third-party dependencies beyond the standard library, with one declared `infrastructure` logging exception (`_logging.py`).
- **Headless plotting**: the analysis script sets `MPLBACKEND=Agg` before importing matplotlib.

## 6.5 Pipeline ordering

The typical analysis order is:

1. `scripts/methods_analysis.py` — compiles every example method, runs all gates, exports worklist/CSV/Mermaid/JSON per method, demonstrates the provenance hash-chain, and writes `../figures/step_counts.png`, printing each output path for manifest collection.

2. `scripts/z_generate_manuscript_variables.py` — reads `manuscript/config.yaml` and the analysis outputs, then resolves every generated variable in `manuscript/*.md`.
3. PDF rendering reads the resolved manuscript tree so figure paths and prose match the analysis that just completed.

### 6.6 Relation to results

| Result (sec. 4) | Producing function (`src/template_methods_paper/methods_dsl/`) | Primary inputs |
|---|---|---|
| Compiled-plan summary | `compile_method()` | `all_example_methods()` |
| Step-count figure | `len(plan.steps)` per method | `output/data/compiled_plans.json` |
| Gate pass tally | `run_all_gates()` | Each example method |
| Determinism check | `compile_method()` called twice | `all_example_methods()` |
| Trust-chain verification | `append_record()` / `verify_chain()` | Demonstration chain in `scripts/methods_analysis.py` |

This table is descriptive documentation only; it is not executed as code during the build.

# 7 Reproducibility

This section explains how to regenerate every artifact in the study from a clean checkout. The exemplar's reproducibility guarantee is structural: each result is produced by a tested function and a thin script, then injected into the manuscript by generated-variable substitution — never transcribed by hand.

## 7.1 How to regenerate everything

From the repository root:

```
# 1. Run the analysis (compiles methods, exports artifacts, writes figure + reports)
uv run python projects/templates/template_methods_paper/scripts/methods_analysis.py

# 2. Run the test suite with the coverage gate
uv run pytest projects/templates/template_methods_paper/tests \
    --cov=projects/templates/template_methods_paper/src --cov-fail-under=90

# 3. Generate and inject manuscript variables
uv run python projects/templates/template_methods_paper/scripts/z_generate_manuscript_variables.py

# 4. Render the manuscript
uv run python scripts/pipeline/stage_03_render.py --project templates/template_methods_paper
```

Or, end to end via the orchestrated pipeline:

```
uv run python scripts/runner/execute_pipeline.py --project templates/template_methods_paper --mode core
```

## 7.2 Generated artifact registry

The analysis script writes the following artifacts under `projects/templates/template_methods_paper/output/`:

| Artifact | Produced by |
|---|---|
| `data/pbspreparation_worklist.md`, `data/pbspreparation_plan.csv`, `data/pbspreparation_graph.mmd`, `data/pbspreparation_plan.json` | `compile_method()` + exporters, for `PBSPreparation` |
| `data/sensorcalibrationsweep_worklist.md`, `data/sensorcalibrationsweep_plan.csv`, `data/sensorcalibrationsweep_graph.mmd`, `data/sensorcalibrationsweep_plan.json` | `compile_method()` + exporters, for `SensorCalibrationSweep` |
| `data/compiled_plans.json` | Per-method plan summary, consumed by `src/template_methods_paper/manuscript_variables.py` |
| `reports/gate_report.json` | `run_all_gates()` tally across both methods |
| `reports/trust_chain_report.json` | `append_record()`/`verify_chain()` demonstration chain |
| `figures/step_counts.png` | Step-count bar chart |
| `data/manuscript_variables.json` | Every generated-variable value, written by `z_generate_manuscript_variables.py` |

The `output/` tree is disposable and regenerated on every run; it is not the source of truth.

## 7.3 Determinism

- `compile_method()` is deterministic by construction: the plan hash is computed from a canonical, sort-keys JSON payload over `method_name`, `method_version`, `target`, and each scheduled step's identifying fields — never from a wall-clock timestamp or a UUID.
- `topological_order()` breaks scheduling ties by ascending `step_id`, so the same `Method` object always yields the same step order across processes and platforms.
- Yes — `src/template_methods_paper/manuscript_variables.py::generate_variables` recompiles every example method twice at manuscript-build time and compares hashes live, so this guarantee is checked on every build, not merely asserted once in a test.

## 7.4 Verification (no hand-transcribed numbers)

Every quantitative claim in sec. 4 is either a generated variable sourced from a live analysis output or registered in `data/claim_ledger.yaml` for evidence-registry validation. The manuscript intentionally does not hand-transcribe volatile values, so prose and artifacts

cannot disagree. Configuration provenance is itself injected: `832fd58172d5b589` is the SHA-256 of `manuscript/config.yaml` at build time, and `2026-10-22T01:20:41Z` records when the variables were generated (honoring `SOURCE_DATE_EPOCH` for byte-reproducible builds).

# 8 Scope, Related Work, and Positioning

This section situates the exemplar and states explicit boundaries. The goal is not to compete with BPL's substantially larger full compiler pipeline [Bota Biosciences, 2026] — with its Lark grammar, robot backend, and hash-chained audit/compliance layer — but to show how a minimal, test-backed subset of BPL's domain-language design fits the template's reproducibility and rendering stack [Peng, 2011], generalized from wet-lab protocols to any controlled procedure.

## 8.1 Domain-specific languages for controlled procedures

Encoding a procedure as a program rather than free text is a long-standing software-engineering pattern: Fowler's treatment of domain-specific languages [Fowler, 2010] frames the general case — a notation restricted to exactly the concepts a domain needs. BPL [Bota Biosciences, 2026] applies this specifically to laboratory protocols, adding biology-native types (reagents, labware, MW-aware concentrations), staged validation, and deterministic compilation to a robot or human execution target. The present manuscript restricts attention to the parts of that design that generalize beyond biology: a controlled vocabulary of step intents, a small dimensional-safety unit system, staged validation gates, and deterministic compilation to a hashed plan.

## 8.2 What this exemplar does not implement (out of scope)

1. **A text grammar and parser.** BPL parses `.bpl` source through a Lark grammar into a typed AST. This exemplar constructs a `Method` directly as frozen Python dataclasses — no concrete syntax, no parser, no AST layer.
2. **Biology-native types.** BPL's unit system includes MW-aware concentration conversions and reagent physical-form metadata (`cas`, `physical_form`). This exemplar's `units.py` implements only the dimensional-safety subset (mass, volume, temperature, time, molar concentration, mass concentration, count, dimensionless) needed to demonstrate the "the type system catches `mL + g`" guarantee — molar and mass concentration are kept as distinct dimensions precisely because no MW-aware conversion exists here to safely mix them.
3. **A robot backend.** BPL lowers intents to Biomek i7 primitives (`aspirate`, `dispense`, `pick_tips`). This exemplar's `Target.AUTOMATED` has no backend-specific lowering stage; it is a scheduling and gate-compatibility concept only.
4. **Cryptographic audit guarantees.** `trust.py`'s hash-chain is deliberately scoped to the same honest boundary BPL itself claims: a consistency check against accidental corruption, not a tamper-proof guarantee against an actor with write access to the entire chain.
5. **SOP-to-DSL generation.** BPL includes an agentic workflow that translates natural-language SOPs into validated `.bpl` programs. This exemplar's worked examples are hand-authored Python, not generated.

## 8.3 What this project proves about the template

The validation and compilation steps here are a deliberately small subset of BPL's. The **non-standard** contribution is procedural: the same tested functions in `src/template_methods_paper/methods_dsl/` back the analysis script, the test suite, and this manuscript, so the compiled-plan table and the figure always refer to the same code. That pattern — and the specific generalization from a biology-only domain language to a domain-neutral one — is what downstream projects should copy, whether the controlled procedure is a wet-lab protocol, an instrument calibration sweep, or a computational pipeline.

## 8.4 Explicit limitations

1. **Two worked examples**: `PBSPreparation` and `SensorCalibrationSweep` exercise every gate-success path but are not a corpus; gate-failure coverage instead lives in `tests/conftest.py`'s dedicated fixtures.
2. **No backend lowering**: `Target` selects a compatibility class, not a concrete execution backend; no robot or simulation runtime exists in this exemplar.
3. **Unit table, not a full unit library**: eight dimensions and a fixed unit table, not a general-purpose dimensional-analysis library like `pint`.
4. **No persistence layer**: `trust.py`'s hash-chain lives in memory for the duration of one script run; this exemplar does not implement durable storage for a real audit trail.

These limitations are intentional: they narrow the surface so that the reproducibility concerns — tested functions, a thin script, and generated-variable-injected prose — remain visible rather than buried under a compiler implementation at BPL's full scale.

# 9 References

Bibliography lives in `manuscript/references.bib` and is read by Pandoc during PDF render. The build pipeline invokes Pandoc with `--natbib`, so every `[@key]` citation in the manuscript is rewritten to the appropriate `\cite{}`/`\citep{}`/`\citet{}` LaTeX command and resolved against the bib file.

To validate that `references.bib` is syntactically clean and contains the required fields per entry type:

```
uv run python -m infrastructure.reference.citation.cli validate \
    projects/templates/template_methods_paper/manuscript/references.bib --strict
```


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